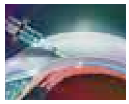
Zebra stripes aren't good landing spots
Researchers proved that Zebra stripes are utilized to prevent effective landing of horse flies!
https://dgit.in/mar19-12



Needle that knows where to go
Researchers developed a highly sensitive intelligent-injector that senses tissue density to properly and safely deliver medication. **https://dgit.in/mar19-25**

a liquid form. Nicotine has an unusual brown colouration, while berberine, colchicine and canadine are orangish. Berberine is found in turmeric. Betaine and sanguinarine both have a red colouration. Most alkaloids have a characteristic bitter taste.

Alkaloids have also been found in some species of fungi and bacteria. The molecules are also present in the animal kingdom, and one of the most common creatures with alkaloids in them are millipedes, some poisonous toads and even the lady bird beetle and some species of moths. The bitter taste is an indication of the poison, and makes the predators avoid these creatures.

There are small amounts of alkaloids present in mammals as well and one of the earliest reported findings was that of alkaloids in goat milk. It was suspected that the origin of the alkaloids was from ingestion of plant matter. Morphine, an alkaloid present in the poppy family of plants, is found in significant quantities within mammals. It is unknown if the origin is from within the mammalian body or from external sources. While scientists know that small amounts of alkaloids appear in mammals, the exact biosynthesis and genetic origin of these compounds is still a matter of research.

Generally, in organisms including plants, fungi, bacteria and animals, alkaloids are derived from amino acids, which are in turn formed from polypeptide sequences derived from sequences encoded within DNA. How the amino acids form into alkaloids, is still not known. Some of the alkaloids found within mammals include dopamine, tryptamine and serotonin. If the organism is suffering from a disease, the levels of these alkaloids rise within the body.

KNOWN FUNCTIONS

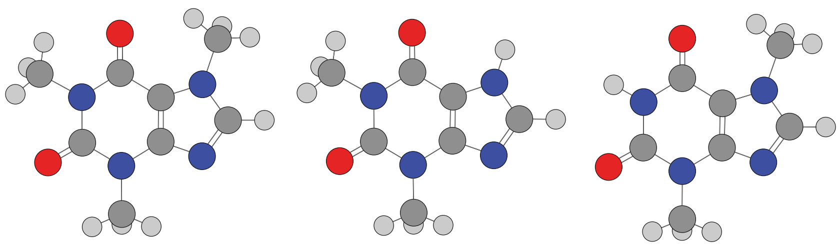
There are two approaches to explain the function of alkaloids, inside and outside the organisms producing them. The simplest explanation is that they are some kind of bioweapon, protecting the organisms from natural predators. However, researchers have used genetic engineering to totally remove alkaloids from some species of plants, and these plants did not survive. The research indicates that alkaloids also serve some kind of function in relation to cellular activity, and are connections to the DNA, enzymes and proteins that perform essential functions for the body. Research also suggest that the alkaloids regulate the growth of the plants, and act as reservoirs of nitrogen. In animals, some alkaloids serve as neurotransmitters. Within the cells of both plants and animals, are structures known as vacuoles which are storage bubbles. Alkaloids within the vacuoles are non toxic, but are toxic outside of them, which indicates that the toxicity of the alkaloids is contingent on the environment which they are in.

Alkaloids have a structure very similar to growth hormones, and there is an old hypothesis that these molecules act like hormones in plants. Various studies have attempted to both prove and disprove this hypothesis, and no scientific consensus has been reached. Unlike animals, the immune systems of plants do not produce antibodies. Instead, they produce phytoalexins, which increase in concentration when there is an infection. There is research to indicate that alkaloids also perform a similar role as phytoalexins. The alkaloids within plants are also antimicrobial, antifungal, antibacterial and even antiparasitic.

MEDICINAL USES

The medicinal uses of alkaloids are varied and many, and have been known since the dawn of civilisation. Alkaloids have historically been used in traditional medicine, but continue to be used in contemporary medicine as well. The alkaloids may be used in their natural form, distilled from plant extracts, or entirely synthesized. Alkaloids are often the starting points of discovering new drugs. One of the earliest active ingredients in plants to be isolated was morphine, in 1803. It is used as a painkiller for chronic and acute pain. Quinine, first isolated in 1820 from the bark of the cinchona tree, is used to treat malaria. Chloroquinine is another antimalarial alkaloid. Ephedrine is used in traditional Chinese medicine, and is currently used to treat low blood pressure, asthma and narcolepsy. Galantamine, which is found in daffodils and some lilies, is used to treat Alzheimer's disease and other memory impairments. Ajmaline, an alkaloid first isolated by Salimuzzaman Siddiqui and named after the Hakim Ajmal Khan, an Unani medical practitioner, is used as an antiarrhythmic to suppress abnormal heart beats.

Several alkaloids, including taxol, vinblastine, vincristine and homoharringtonine act as anticancer agents. These alkaloids are used in combinations with other drugs to treat many types of cancer. Alkaloids are expected to be the starting points for the continued discovery of new drugs, for the benefit of man. Alkaloids are not always beneficial, and overuse or abuse of alkaloids can lead to problems in humans, including narcosis and addiction.



Caffeine   Theophylline   Theobromine

Shown here are three common alkaloids. The light grey circles are hydrogen, the dark grey circles are carbon, the blue circles represent nitrogen, and the red circles are for oxygen.

More detailed information on the subject can be found in the book, Alkaloids - Secrets of Life by Tadeusz Aniszewski.